

Sales Forecasting Model Enhancement

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Shift Digital

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Introduction

Shift Digital's forecasting platform generates thousands of store-model- day level predictions per client, but the system struggles with highly seasonal businesses and extremely sparse sales data. These challenges may lead to unstable predictions and inconsistent accuracy across different client types, and program budgeting, marketing optimization, and dealer performance insights.

- This study addresses three core research questions.
- How do time-series forecasting models behave under seasonal versus non-seasonal demand structures, and which model characteristics determine their performance across these environments?
 - Which forecasting methods are most effective for sparse, low-frequency retail time series, and through what mechanisms do they reduce error variability and instability?
 - To what extent can pattern-specific forecasting strategies reduce overall forecasting error in operational contexts, and which business types benefit most from segmentation?

In this project, we address these issues by classifying each store-item series into meaningful demand patterns and benchmarking forecasting models tailored to those patterns. Our goal is to propose a practical, pattern-aware forecasting framework that improves accuracy for challenging dealership segments and strengthens decision support across Shift's analytics workflow.

Data

The dataset used in this study contains sales data from Shift Digital's internal sales and marketing system. It includes thousands of store-model-date level records across multiple years, which were aggregated to monthly frequency. Store and item IDs are anonymized GUIDs in the dataset, which means no geographic, demographic, or product category information is available.

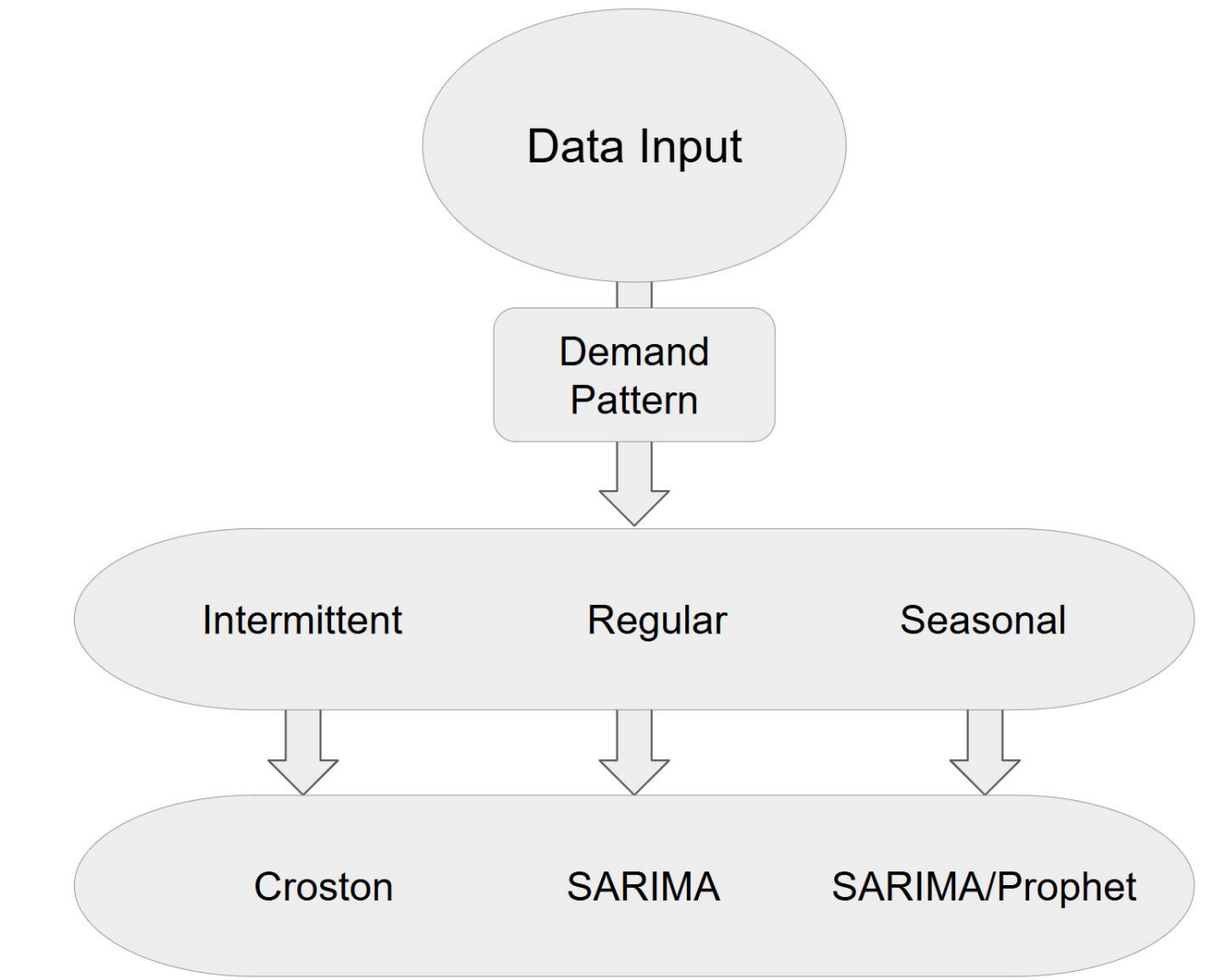
The key variables include Date (transaction timestamp), StoreId (unique store identifier), ItemId (vehicle model), and Sales (units sold). There are 185,292 rows, and 4 columns. The dataset records time range from 2023/07/01 to 2025/08/01. There are 750 stores, and 84 items in total as shown in table. Most series exhibit extremely low transaction volume, with over 75% of observations recording a single unit sold. This sparsity creates forecasting instability and motivates a structure-based approach rather than a one-model-fits-all framework.

To characterize the structural behavior of each time series, the dataset was further processed to compute a set of diagnostic indicators, including zero-share, the average inter-demand interval (ADI), and autocorrelation at the twelve-month lag (ACF12). These diagnostics enable classification into intermittent, regular, and seasonal patterns, forming the basis for model selection and performance evaluation.

Variable	Description	Type	# Unique
Date	Transaction timestamp	Datetime	762
StoreId	Unique store identifier	Categorical	750
ItemId	Vehicle model identifier	Categorical	84
Sales	Number of units sold	Numeric	31

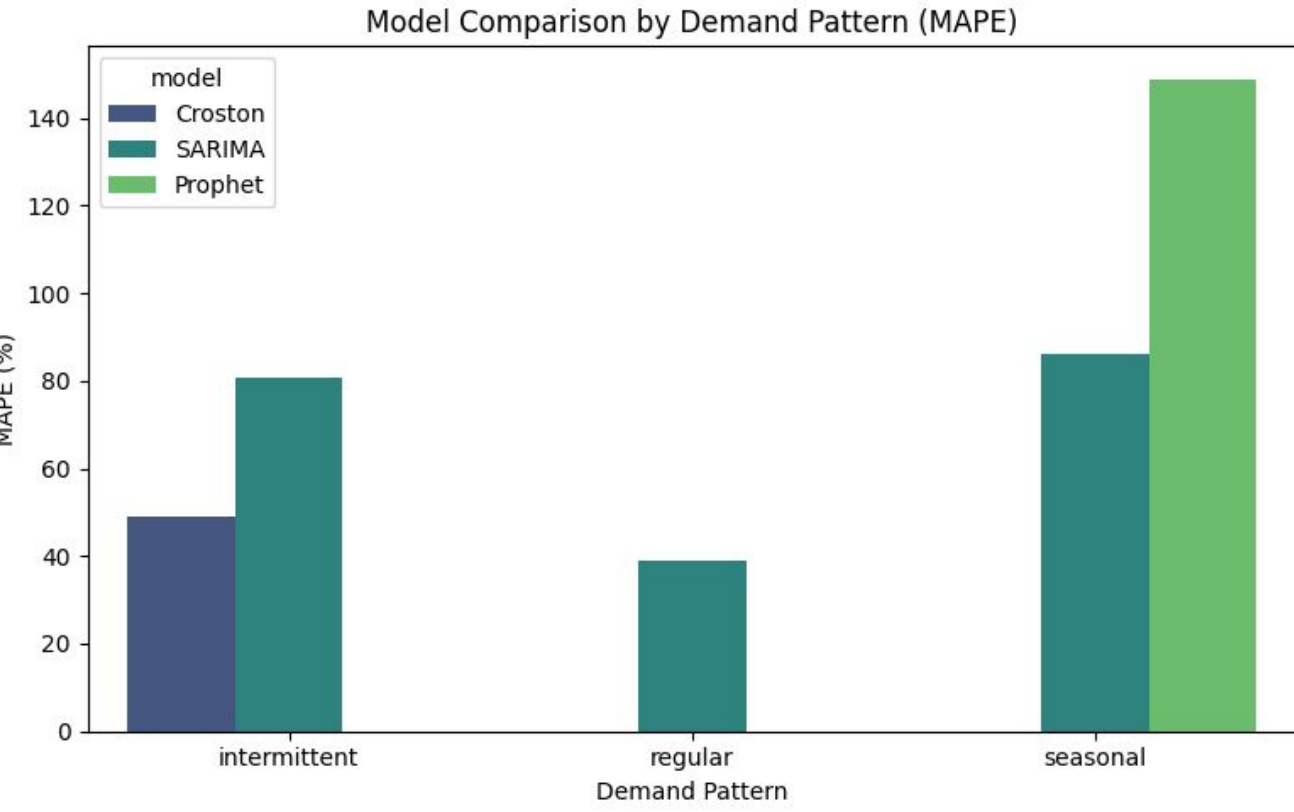
Methodology

The forecasting approach adopts a pattern-specific modeling framework in which each store-item time series is matched with a model suited to its underlying demand structure. After demand patterns were identified in the data preprocessing stage, intermittent series were forecast using Croston's method, which separately estimates demand size and the interval between non-zero sales and is well suited for sparse, zero-inflated processes. Regular and seasonal series were modeled using SARIMA, whose autoregressive and moving-average components capture stable temporal dynamics and moderate seasonal behavior. Prophet was also applied to seasonal series to evaluate the performance of a flexible decomposition-based model under limited seasonal depth. Forecast accuracy was assessed using a comprehensive set of metrics, including MAPE, RMSE, MAE, sMAPE, Bias, and MASE, and model robustness was evaluated through distributional error analysis and win-count comparisons. Only methods that produced valid and stable forecasts for the corresponding pattern were included in the final evaluation.



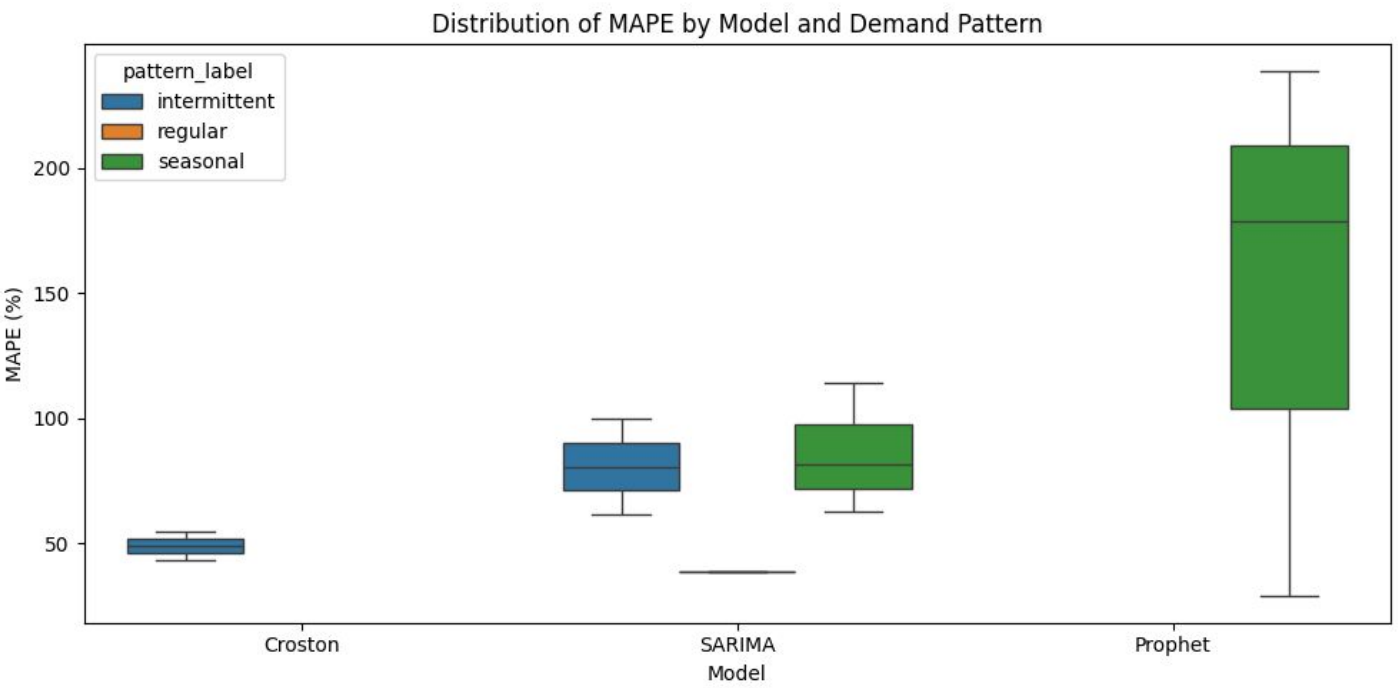
Results

To evaluate forecasting performance across demand patterns, we compared Croston, SARIMA, and Prophet using six accuracy metrics. Mean MAPE provides a high-level view of how model accuracy differs across intermittent, regular, and seasonal series. Results show substantial accuracy variation across patterns, supporting the need for pattern-specific modeling.



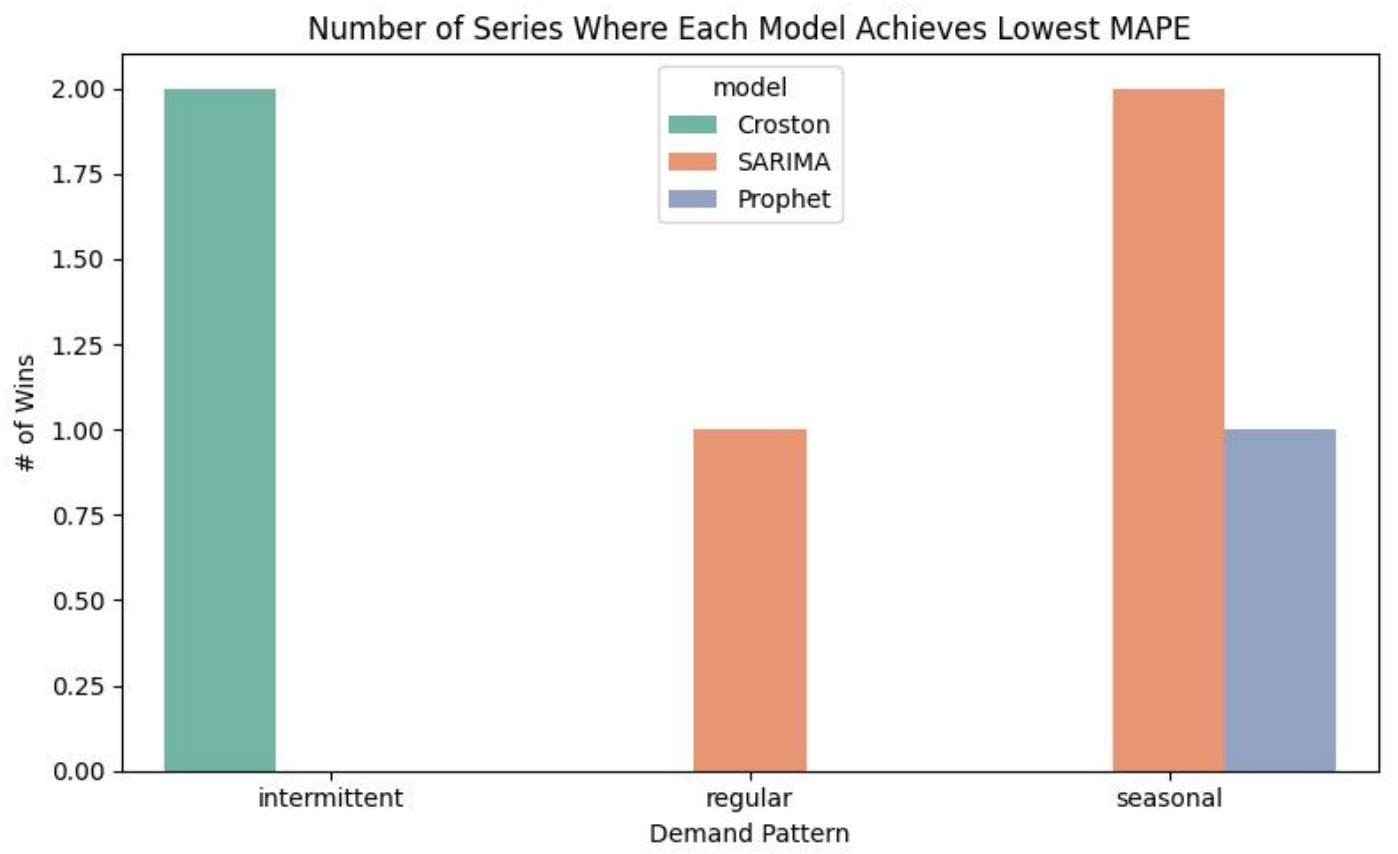
Mean MAPE by model and pattern. Croston performs best for intermittent demand; SARIMA for regular and seasonal.

While mean accuracy summarizes central performance, the distribution of errors reveals important differences in model stability. Boxplots show that Croston and SARIMA produce more consistent errors, whereas Prophet exhibits higher dispersion for seasonal series.

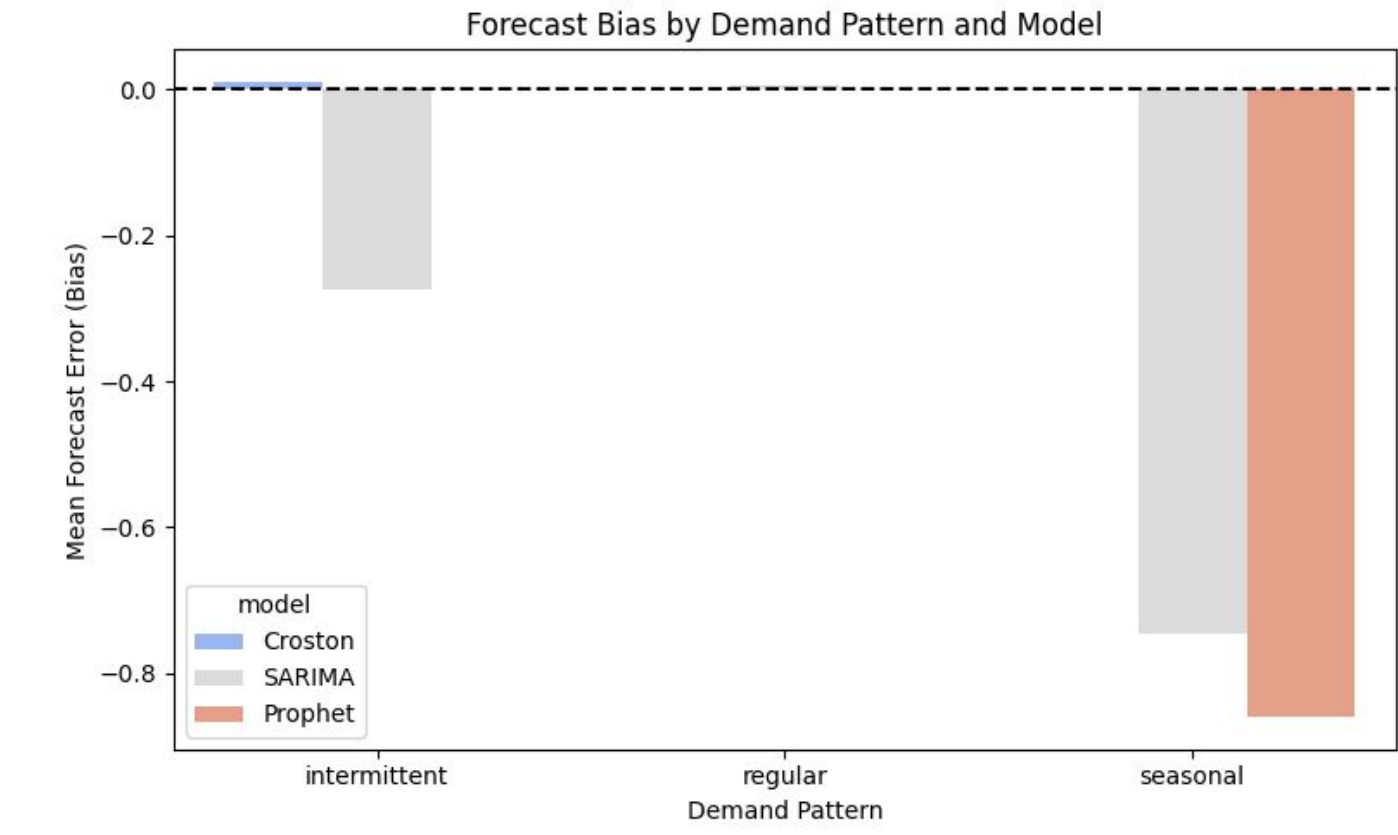


MAPE distributions. Croston is more stable for intermittent demand; SARIMA has tight error ranges for regular and seasonal series.

We also compared models by counting how often each achieved the lowest MAPE across series. This provides a discrete measure of comparative dominance within each pattern.



Number of store-item series for which each model achieved the lowest error.



Mean forecast bias across models. Prophet exhibits largest under-forecasting for seasonal demand.

Bias patterns reveal systematic over- or under-forecasting, which is critical for operational use. Prophet substantially under-forecasts seasonal peaks, whereas Croston and SARIMA produce near-zero bias for intermittent and regular demand.

Summary of Forecasting Performance Across All Metrics

Demand Pattern	Model	MAPE	RMSE	MAE	sMAPE	Bias	MASE
intermittent	Croston	48.864	0.522	0.519	134.179	0.011	1.220
intermittent	SARIMA	80.633	1.237	1.069	159.164	-0.274	2.214
regular	SARIMA	38.848	0.482	0.454	48.427	0.005	0.349
seasonal	Prophet	148.904	3.902	3.344	101.285	-0.861	2.549
seasonal	SARIMA	86.126	2.439	2.129	118.171	-0.748	1.526

Forecasting performance across all models and demand patterns using six error metrics (MAPE, RMSE, MAE, sMAPE, Bias, MASE). Lower values indicate better performance, except Bias (values near zero indicate minimal over/underprediction).

Conclusion

This project shows that forecasting accuracy at the store-model level cannot be improved with a single universal model. Instead, demand patterns drive performance, and model selection must adapt to how each store actually sells.

For intermittent demand, Croston consistently provides the most reliable forecasts because it separates demand occurrence from demand size, especially for patterns with long stretches of zero sales. For regular demand, SARIMA delivers the strongest and most stable results. For seasonal demand, SARIMA again outperforms alternatives, but Prophet struggles under short and irregular seasonal histories.

These findings highlight a clear business implication that Shift Digital can meaningfully improve forecast accuracy by automatically detecting demand patterns and routing each series to the appropriate model, rather than applying a one-model-fits-all strategy. This approach is simple to operationalize and can be integrated into the existing forecasting workflow.

Going forward, richer store attributes (e.g., marketing spend, inventory availability, regional demand factors) and longer sales histories would enable machine-learning and hybrid methods to play a larger role. But with the current data structure, a pattern-specific statistical framework provides the highest and most stable value.

Acknowledgements

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